

ANALYSTLAB AFRICA
Experience Lab Internship Programme



HealthConnect Clinic
Week 7 Project Summary

By
Ogheneughwe Godwin AKISE

Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance
and Healthcare Support Using Data and AI
Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 7 Submission – Model Testing, Error Analysis & Refinement

This Week 7 Project Summary accompanies the Data Science track's HealthConnect Model Testing & Refinement Report. It is deliberately kept concise: it evaluates this week's testing work against what was planned, and highlights what was found, decided and changed, without repeating the full test-by-test tables and figures already documented in the main report.

1. What you planned to test

Week 7 was planned around the five limitations the Week 6 report itself had flagged as untested: whether the Week 6 candidate's small reported improvement over the Week 5 baseline was stable across validation samples; whether its errors concentrated in identifiable patient or appointment groups; whether its engineered features remained stable; whether the provisional 0.45 operating threshold represented a defensible trade-off; and whether its cross-track requirements had been end-to-end validated. An eight-test plan (T01–T08) was defined to answer these questions, covering cross-validation stability, a fair baseline-vs-candidate comparison, error and segment analysis, feature stability, overfitting, threshold validation, and mandatory cross-track validation with Data Analytics.

2. What you actually tested

All eight planned tests were executed and are reported in full in the main deliverable. Beyond the original plan, two additional pieces of work were carried out once testing was under way: a targeted calendar-feature ablation, tested as a genuine refinement rather than assumed, and a verification pass comparing every test's narrative interpretation against its own executed output, which surfaced two issues not anticipated at the planning stage (Section 4).

3. The most important testing results

- T01 (stability): the candidate's cross-validated ROC-AUC ranges from 0.6519 to 0.6959 across five folds (mean 0.6748) — the Week 6 single-split figure of 0.6890 sits inside this normal range rather than being a fixed, reproducible number.
- T02 (baseline comparison): on a fair, fold-matched comparison, the candidate beats the Week 5 baseline's ROC-AUC in 3 of 5 folds and PR-AUC in 2 of 5 folds, but the mean difference across all five folds is slightly negative for both metrics (-0.0008 ROC-AUC, -0.0006 PR-AUC) — the Week 6 reported improvement is not robust under repeated validation.
- T07 (threshold): the provisional 0.45 threshold avoids 188 missed no-shows at the cost of 185 additional unnecessary flags, a trade-off of roughly 0.98 extra false positives per false negative avoided.
- T08 (cross-track): the Data Analytics track's lead-time and prior-no-show-history findings were directionally reproduced in the Data Science modelling sample once segment definitions were harmonised — this cross-track test passed.

4. Issues or weaknesses identified

- The candidate's errors are concentrated, not random: short-lead-time bookings have very high false-negative rates (up to 0.876 for 0–7 day bookings), while long-lead-time and distant bookings have very high false-positive rates (up to 0.910 for 31–60 day bookings).

- Two calendar-derived features (`appointment_day_of_year`, `appointment_month`) showed materially higher coefficient variability across folds than the Week 6 engineered interaction features, prompting the ablation test described in Section 5.
- A feature-reconstruction gap was initially found in the Week 7 testing notebook: its helper function rebuilt only two of the Week 6 candidate's three engineered interaction features, omitting `history_interaction`, so early Week 7 testing ran on a 37-feature reconstruction rather than the full 38-feature Week 6 specification. This was corrected during the week, and all figures in the final main report and this summary reflect the complete 38-feature reconstruction.
- A narrative/execution inconsistency was found in the notebook's own T02 interpretation (and the records that repeated it): it stated a 1-in-5 fold win rate for the candidate, where the test's own executed output and underlying comparison table show 3-in-5. This, and several related instances of interpretation text lagging behind re-executed cell output elsewhere in the notebook, were tracked and corrected (Section 11).

5. Improvements/refinements made

Rather than removing the two higher-variability calendar features on variability alone, a targeted calendar-feature ablation was tested against a pre-declared tolerance of 0.002 on the full 38-feature reconstruction: removing `appointment_day_of_year` and `appointment_month` changed mean five-fold ROC-AUC, PR-AUC and Recall@0.45 by amounts that, once the feature-reconstruction gap in Section 4 was corrected, no longer all held within the pre-declared tolerance. The ablation was therefore rejected, and the complete 38-feature Week 6 specification was retained as the Week 7 refined model rather than the simplified one. The 0.45 threshold was retained unchanged, since no test produced evidence to revise it beyond the trade-off already quantified.

6. Retesting results

The retained 38-feature model was retrained on the full Week 5 training partition and evaluated once on the project's held-out test set: Accuracy 0.6263, Precision 0.6066, Recall 0.7184, F1 0.6578, F2 0.6929, ROC-AUC 0.6890, PR-AUC 0.6717 at the 0.45 threshold (258 true negatives, 225 false positives, 136 false negatives, 347 true positives), consistent with the cross-validated range obtained during T01. The retest confirms the Section 5 refinement decision: keeping the full feature specification performs in line with expectations and introduces no new instability.

7. Which track(s) you collaborated with

Data Analytics Track, as the mandatory Week 7 HC-POD cross-track testing partner.

8. What was tested collaboratively

The Data Analytics Week 7 report's independently validated findings — an overall no-show rate of 51.15% across 4,737 eligible appointments, a rising no-show rate with booking lead time, and a rising no-show rate with prior no-show history — were tested against the Data Science modelling sample. This surfaced a genuine definition mismatch: the two tracks had used different booking-lead-time bands. The Data Science segment test was rebuilt using the exact Analytics bands (0–7, 8–21, 22–40, 41–60 days) and history groups (0, 1, 2+ prior no-shows) and re-evaluated.

9. What changed as a result

Data Science's segment-testing definitions are now explicit and harmonised with the Data Analytics Week 7 bands, removing the risk of the two tracks describing the same business relationship with silently different cut-points. Both the lead-time and prior-no-show-history directional findings were reproduced in the modelling sample once harmonised, with the combined high-risk segment (41–60 days, 2+ prior no-shows) showing an 84.5% no-show rate in the modelling sample against the Analytics benchmark of 83.33%, consistent given the two tracks draw on different populations.

10. Key findings or validation outcomes

- The Week 6 candidate is directionally stable and shows no material overfitting (mean train/validation ROC-AUC gap of 0.0166), but its improvement over the Week 5 baseline is not a reliably reproducible gain.
- The candidate's error profile is systematically structured around booking lead time and, to a lesser extent, distance to clinic — a genuinely actionable finding for HealthConnect's outreach design, not just a diagnostic curiosity.
- The calendar-feature ablation is a validated, evidence-based refinement decision in the opposite direction from the one first suspected: once tested correctly on the full 38-feature reconstruction, the two calendar features did not clear the pre-declared tolerance for removal, so the complete Week 6 specification was retained rather than simplified.
- The mandatory Data Analytics cross-track dependency has passed on both tested directional findings, closing one specific integration risk ahead of Week 8.

11. Major challenges encountered

- Reconciling a materially more cautious T02 result (improvement not robust under cross-validation) against the more favourable single-split figure reported in Week 6, and deciding to report both rather than only the more favourable one.
- The processed Week 5 training matrices still do not retain patient_id, so patient-grouped cross-validation remains unavailable in Week 7, as it was in Week 6; StratifiedKFold on individual appointments was again used as the practical alternative.
- A genuine computation bug was found and corrected in the feature-stability test's variability table: its standard-deviation column had been computed over six values (the five fold coefficients plus an already-appended mean) instead of five, understating true coefficient variability — the figures reported throughout the final deliverables use the corrected five-fold calculation.
- Verifying every test's narrative interpretation against its own executed output, rather than accepting the notebook's markdown commentary at face value, took several iterative passes: it is what surfaced the feature-reconstruction gap, the T02 narrative/execution mismatch, the std-computation bug, and a handful of smaller interpretation cells that had not been regenerated after the feature-reconstruction fix — all now resolved and consistent with executed output.

12. Important decisions made

1. To report the verified T02 result (3/5 ROC-AUC wins, 2/5 PR-AUC wins, mean differences of -0.0008 and -0.0006) rather than the Week 7 notebook's own, internally inconsistent narrative figures, once the discrepancy was confirmed against the test's executed output.
2. To test the two higher-variability calendar features via a pre-declared, tolerance-based ablation rather than removing them on coefficient variability alone, and to honour that rule's outcome — rejecting the ablation — once the feature-reconstruction gap was fixed and the test re-run correctly, rather than keeping the earlier, incorrectly-derived accept decision.
3. To retain the 0.45 operating threshold as provisional, rather than revising it, since T07 quantified its trade-off precisely but no HealthConnect stakeholder tolerance for that trade-off has yet been confirmed.
4. To harmonise Data Science's segment definitions to the Data Analytics Week 7 bands for the T08 cross-track retest, rather than treating a simple exchange of reports as sufficient collaboration evidence.

13. Remaining limitations

- Patient-grouped cross-validation remains unavailable, since patient_id is not retained in the processed modelling matrices.
- Preprocessing was fitted once, prior to the Week 7 fold split, rather than within a fully encapsulated per-fold pipeline, carrying a small residual leakage risk.
- The held-out test set used for retesting (Section 6) has already been evaluated in earlier weeks and is not a wholly unseen external validation set.

14. Remaining issues or dependencies

- Prediction-level appointment-ID traceability, raised in the Data Analytics Week 7 report, is unresolved and was explicitly not claimed as resolved by the T08 cross-track activity.
- A missing-distance data discrepancy flagged by Data Analytics (86 eligible records with missing distance data in their source versus 39 in the working dataframe used for this test) has not yet been investigated by this track.
- ML Engineering pipeline validation of the final model artefact, feature schema and threshold has not yet been completed.
- A formal HealthConnect stakeholder decision on the acceptable recall/false-positive trade-off at the 0.45 threshold is still outstanding.

15. Your contribution to the overall HealthConnect project

Week 7 moves the HealthConnect no-show-prediction workstream from an unverified, single-split improvement claim to a tested, honestly qualified candidate model: its genuine strengths (directional stability, no material overfitting, a feature set tested rather than assumed) and genuine weaknesses (a non-robust improvement over baseline, concentrated segment-level errors, an unresolved operational threshold trade-off) are now both documented with executed evidence rather than narrative alone. Holding every reported figure to its own executed cell

output — rather than the notebook's narrative commentary — also corrected a feature-reconstruction gap, a genuine standard-deviation computation bug, and several stale interpretation passages before they could propagate into the final record, leaving the retained 38-feature model and its retest metrics fully reconciled between the underlying notebook and both Week 7 deliverables. The completed Data Analytics cross-track validation further strengthens the project's overall evidentiary base by independently reproducing two of Data Analytics' key findings inside the modelling pipeline itself, rather than merely referencing them.

16. What must be completed before Week 8

1. Freeze and version the final model artefact (the retained 38-feature specification), its feature schema and the 0.45 threshold.
2. Obtain a formal HealthConnect stakeholder decision on the 0.45 threshold's recall/false-positive trade-off.
3. Resolve the two outstanding Data Analytics cross-track items: appointment-ID traceability and the missing-distance discrepancy.
4. Complete ML Engineering pipeline validation of the final model artefact.
5. Prepare Week 8 presentation evidence summarising the testing, refinement and cross-track validation record established this week.

References

- AnalystLab Africa. (2026). *HealthConnect Clinic: Model improvement & validation report*. Week 6 submission.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Model testing & refinement report*. Week 7 submission.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Data Analytics track Week 7 testing and validation report*. Week 7 submission.
- Dantas, L. F., Fleck, J. L., Cyrino Oliveira, F. L., & Hamacher, S. (2018). No-shows in appointment scheduling – a systematic literature review. *Health Policy*, 122(4), 412–421.
<https://doi.org/10.1016/j.healthpol.2018.02.002>
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-0716-1418-1>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
<https://scikit-learn.org/stable/modules/preprocessing.html>
- Saeb, S., Lonini, L., Jayaraman, A., Mohr, D. C., & Kording, K. P. (2017). The need to approximate the use-case in clinical machine learning. *GigaScience*, 6(5), gix019.
<https://doi.org/10.1093/gigascience/gix019>